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$$I = \int_0^1 \frac{x^2 \operatorname{Bgtan}(x)}{1+x^2} dx$$
$$= \int_0^1 \frac{\left(x^2+1-1\right) \operatorname{Bgtan}(x)}{1+x^2} dx$$
$$= \int_0^1 \left(\operatorname{Bgtan}(x) - \frac{\operatorname{Bgtan}(x)}{1+x^2}\right) dx$$
$$I_1 = \int_0^1 \operatorname{Bgtan}(x) dx$$
$$= [x \cdot \operatorname{Bgtan}(x)]_0^1 - \int_0^1 \frac{x}{1+x^2} dx$$
$$= \frac{\pi}{4} - \int_0^1 \frac{x}{1+x^2} dx$$
Stel $t = 1 + x^2$
$$\Rightarrow dt = 2xdx$$
$$\Rightarrow \frac{1}{2} dt = xdx$$
$$I_1 = \frac{\pi}{4} - \frac{1}{2} \int_1^2 \frac{1}{t} dt$$
$$= \frac{\pi}{4} - \frac{\ln 2}{2}$$
$$I_2 = \int_0^1 \frac{\operatorname{Bgtan}(x)}{1+x^2} dx$$
Stel $t = \operatorname{Bgtan}(x)$
$$\Rightarrow dt = \frac{1}{1+x^2} dx$$
$$I_2 = \int_0^{\frac{\pi}{4}} t dt = \frac{\pi^2}{32}$$
$$I = \frac{\pi}{4} - \frac{\ln 2}{2} - \frac{\pi^2}{32}$$

References